Carlos Toledo

Independent research engineer
exact verification of machine-generated claims · benchmark and grader audits · verifiers that cannot be gamed

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I check what AI systems and benchmarks claim — in exact arithmetic, with code that shares nothing with the claimant — and publish each verdict as a re-runnable certificate: certified, refuted with a witness, or refused with the reason.

131	10	3,582 / 4,000	9.75×10^{15}
CLAIMS DECIDED	ASTERISKS REMOVED	WRONG ANSWERS A TOLERANCE GRADER ADMITS	PAIRS OF POINTS DECIDED EXACTLY
since August 2026, each with the record that decided it	from Tao's optimization-constants registry, on our replays	a certificate grader admits none	the Qiushi Engine's nineteen kissing bounds



The register (certs/claims-ledger.json), read when this page was built: decided claims by day; a solid step, because every count is exact.

BENCHMARK AND RELEASE AUDITS

2026 · all public

OpenAI's math release — 719 machine-generated manuscripts and 416 Lean challenges, audited at a pinned commit · report

- Every proof re-run under a second, independent kernel (nanoda, written in Rust): 398 of 416 accepted by both kernels and none rejected; the other 18 exceeded the runners' memory or time.
- Every formal statement read against the claim it carries: 34 of 242 formalized families state a different or weaker theorem than their headline, with nothing saying so; 14 gaps published, each pinned to file and line; a kernel fix posted upstream.

Two claimed refutations of the Exponential Time Hypothesis (OpenAI, Anthropic), replayed · report

- Both proofs accepted by two kernels; the statements read against the textbook ETH by two independent readers; the textbook ETH written once in Lean and each lab's statement proved to refute it in its own machine model.

The graders of mathematical benchmarks, measured against 4,000 provably wrong answers · report

- Absolute-tolerance rules accept 3,582 of them; a certificate grader accepts none and rejects nothing right.

METR's time-horizon estimator, re-derived exactly · report

- 44 of 44 fits certified over METR's public runs; the post-2023 doubling time re-derives to between 128.740 and 128.741 days, against the printed 128.744.

GSM8K's answer key, every arithmetic step re-decided · report

- 8,792 keys (test and train); 4,282 annotated steps, all exact; 26 printed steps wrong, 7 of them leaving the key's answer underived.

INDEPENDENT CERTIFICATION OF MACHINE-GENERATED MATHEMATICS

2026

The Qiushi Engine's kissing-number bounds · report

- 19 of 19 announced bounds witnessed and every pair of 9.75×10^{15} decided exactly; the claimant published its point sets at the audit's request and credits the check in its README. Also the first independent certification of $K(18) \geq 8{,}358$ (34,923,903 pairs, 0 violations).

Registries of AI-generated certificates: Terence Tao's optimization-constants registry removed 10 "verification minimal" asterisks on our exact replays, in four merged pull requests, and cites them as [T2026] (report); its PR queue 9 certified, 1 refused (report); AlphaEvolve's community certificates 70 certified, 2 refused (report); Written on the Wall II 5 of 5.

Lean certificates read against their statements: OpenAI's Navier–Stokes blowup, one clause of the theorem missing at the announced commit and proved at the next (report); the Qiushi Engine's disproof of Bandeira's 0.2(a), the finite core re-decided and two gaps named (report).

Theorems that calibrate the instruments: $\lambda(4)$ for Chowla's cosine problem, re-certified by a third party with no shared code (report); $\delta_3 = 117/2192$; bounds on Erdős #290 and #1038 (Zenodo).

FINANCIAL MODEL RE-PERFORMANCE — October 2026

PU Verificado — every Tesouro Direto price since 2004 re-decided from the Treasury's own rule table · report

- 145,679 prices (2004–2026): 132,538 reproduced to the centavo, 12,791 consistent with the printed rate's precision, 289 off with the exact value and the difference, 61 needing data the publisher kept; none refused.
- Two deciders sharing no code; conventions measured, never assumed: settlement on B3's calendar, centavos truncated.

RL ENVIRONMENTS, VERIFIERS AND THE VISUAL GRAMMAR — 2026

Three reward designs, one environment, on Prime Intellect's Hosted Training — a pre-registered GRPO study: an answer key and an exact grader both extinguished abstention within ~10 steps, a ternary grader made the model abstain on everything. On the Environments Hub: `lattice-claims`, `blind-spot`, `break-the-grader`.

WARRANT, a visual grammar in which every mark says what it rests on · the rules

- The stroke says how a number was made (decided, computed, chosen, refused), words carry the verdict, a ladder how far each premise was checked; the site refuses to build a page that breaks a rule and re-reads every number from its record.

Writing: cert-machine: independent exact certification of machine-generated mathematics (method paper, draft, not peer-reviewed) · PDF. Outside re-derivations of the verdicts, all concordant, are on the record with their authors' names (report).

HOW I WORK

Verifier first: a check that has never gone red is decorative, so every battery carries planted forgeries. Pre-registration before any spend, deviations dated, negative results reported; every number read from a record, this page's included.

Research: Python (standard library, Inspect, verifiers, pytest), JavaScript (BigInt exact arithmetic, headless-Chrome gates), Lean 4 (statements, builds, axioms, two kernels), SAT, LaTeX; GPU campaigns on RunPod; Hosted Training.

Product: design systems and the visual grammar, Figma, Webflow (CMS, GSAP), API integrations, SEO, accessibility.

EXPERIENCE

cert-machine — Independent research engineer — 2026–present

The audits, the re-performance, the RL work and the grammar above; open source, one author.

Finsweet — Senior Webflow Developer — 2020–present

End-to-end delivery of Webflow sites for global clients, from concept to deployment: CMS architecture, custom logic, advanced animation.

Beacon / Embraer — Webflow Developer & API Integration Specialist — 2021–2023

Migrated complex WordPress systems to scalable Webflow stacks; built enterprise frontends integrated with internal APIs and third-party services.

Filmelier — Product Designer — 2019–2020

Editorial UX and a Figma-to-Webflow production pipeline; SEO and accessibility.

Brainn, Movile, ED Interactive, Make ID, TAG, LabCom — Designer — 2010–2019

Digital design, branding, UX/UI and product work.

EDUCATION

Universidade Estadual Paulista (UNESP) — Bachelor's in Industrial & Graphic Design — 2003–2009

SENAI São Paulo — BTech in Graphic and Printing Equipment Operation — 2004–2006